

Multimodal Attention Networks for Human Activity Recognition From Earable Devices

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ABSTRACT

Earables (a.k.a ear-worn wearable devices) are gaining traction in the wearables ecosystem for monitoring user health. Human activity recognition (HAR) is a promising use case of earables due to their placement on the head and the combination of sensors. In this paper, we explore using multimodal attention-based neural networks for HAR from the ear. Attention networks have had a large impact on other disciplines' machine learning tasks and we believe they present opportunities in HAR from earable data. Different methods of utilising attention mechanisms in the literature are discussed as well as the benefits and challenges of using such networks in the context of HAR on real systems.

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1 INTRODUCTION

Earables are emerging as wearable user monitoring device [4]. These can contain multiple sensors which may be harnessed to perform, among other things, human activity recognition (HAR) [7]. Being mounted on the head, earables can extract information about movement of the head as well as the net movement of the body. Like with other wearable data, deep learning shows state-of-the-art results on earable data. This paper explores using an attention mechanism as part of a network for HAR on earable data for their improved performance, robustness and interpretability over other deep learning methods.

Human activity recognition is one of the most fundamental tasks in mobile sensing. It has applications in monitoring wellbeing and healthcare. HAR from wearables can be regarded as a time-series classification task, a long-standing problem with applications to healthcare and finance. In the past, this has been performed with traditional machine learning classifiers [12]. Normally, Inertial Measurement Unit (IMU) data is the main source of data for classification, but other data streams such as heart rate and body temperature can be used for HAR. Recently, deep learning techniques have been

used to make these classifications and extract features in the same pipeline. This can incorporate multiple data streams more easily, and thus increase the robustness of the model [11]. As a lot of the deep learning models used for HAR previously have been inspired by advances in vision like the Convolutional Neural Network (CNN) or sequence modelling such as Recurrent Neural Networks (RNN), we argue that the latest and state-of-the-art technique, attention mechanisms, should be employed for HAR too and that the benefits of doing so outweigh the challenges.

Attention mechanisms [1] are designed to mimic human cognitive attention in a computer by learning what parts of the data are most relevant to each other and, therefore, finding the important features. Neural networks based on attention mechanisms have improved performance on many tasks in deep learning [10]. The transformer model architecture was proposed in 2017 [15] and led to the widespread use of attention-based models. The transformer is a sequence-to-sequence model, of which the main benefit is that it acts over the whole input sequence with an attention mechanism rather than having a limited window such as previous state-of-the-art such as RNNs. The transformer has led to successful large language models such as the Bidirectional Encoder Representations from Transformer model (BERT) [2] and its many derivatives. Besides its use in natural language processing, the transformer and attention mechanisms have led to the development of the vision transformer [3], which achieves state-of-the-art performance on computer vision tasks.

2 ATTENTION NETWORKS: A PRIMER

An attention network contains at least one layer that has learned parameters for the relevance of different inputs relative to the other inputs at each position. This relevance is calculated with a series of matrix multiplications for a query given certain keys and values. For sequence modelling, for example in translation, the input sequence is turned into a series of tokens with a positional embedding as the attention layer inputs. Transformers are an example of a successful purely attention-based network.

3 OPPORTUNITIES OF ATTENTION ON HAR

Opportunity 1: Performance, robustness and versatile designs of attention layers. Using attention networks for HAR from earable data gives improved performance as well as the robustness of using multimodal data such as IMU, audio, PPG and other sensors. An example is the combination of attention and convolution exploited for multimodal timeseries by Tripath et al. [14], which outperforms other deep learning methods on HAR datasets. Additionally, a promising method that resolves where to focus and what to focus on was proposed by Gao et al. [5]. They use both temporal



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and cross-channel attention layers in a Dual Attention network. This network was specifically developed for HAR tasks and shows an increased performance over other deep learning methods on HAR datasets. This combination of different attention mechanisms allows the network to outperform other attention-based networks while remaining end-to-end trainable, unlike some other networks.

Opportunity 2: Multitask generalisation. Another way to exploit the attention mechanism is with its generalisability. These networks can be used to perform multiple tasks on the same data streams. This is particularly relevant to HAR where we may desire information such as gait and posture. Liu et al. developed a multi-task attention-based model that can run on-device [9] showing the possibility for this model to generalise and still be lightweight.

Opportunity 3: Visualising the attention. The interpretability of the model is also a benefit of attention networks, since it involves visualising the attention weights [13]. This has particular importance when applied to healthcare where HAR can be used. Gao et al. also discuss visualisation of channel importance by looking at attention weights [5]. This has applications in reducing the model complexity by not including data streams or features deemed to be less relevant, this could be important to running a model on-device in real-time.

4 CHALLENGES

In the previous section, we discussed the opportunities presented by applying attention networks to earable-based HAR. However, to achieve those, there are a number of challenges researchers have to face to implement these networks onto real systems. This section will reflect on these.

Challenge 1: Keeping the model lightweight. While developing the model, the end use case of running on a mobile device must be kept in mind. This means keeping the model lightweight enough for the target device. Attention models can have higher parameter counts than other deep learning models [16] so an increase in performance must be weighed against a loss in on-device inference time. Liu et al. [9] use a model with temporal shift attention and convolution layers that runs on-device using more efficient convolutions. Established techniques for reducing model size such as quantisation and pruning [8] are important for keeping the model lightweight. Additionally, hardware designed for edge on-device inference is being developed, bringing increased performance in mobile deep learning [17].

Challenge 2: "Data Hungry" models. Attention-based models can be described as "data hungry" and require larger datasets to train compared to other deep learning models [6]. This could reduce the performance of HAR from earables compared to models that are conventionally employed if sufficiently large datasets are not available. One solution to this problem could be to use larger datasets from other wearable devices and implement transfer learning, where a model trained on a larger dataset can learn the characteristics of the sensor data and then be retrained on a smaller, earable and task-specific dataset.

Challenge 3: Adapting techniques from other ML domains and lack of established frameworks. Implementing transformers and attention layers for domains such as natural language and vision is aided by well-developed frameworks. But for timeseries

applications, these frameworks are in their infancy, which can make developing and experimenting with models time consuming. However, there are emerging frameworks, since this is an active area of research and timeseries tasks are well supported for other networks such as CNNs and RNNs. These can be adapted for attention layers.

5 OUTLOOK

Attention networks and transformers have revolutionised natural language and vision fields giving state-of-the-art performance. We have argued the opportunities presented by attention networks for HAR outweigh the challenges of implementing the models on real systems. Researchers will need to focus on keeping the models lightweight enough for mobile devices, the datasets used, as well as schemes to adapt techniques from other ML domains.

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